

Structure-Based 3D Drug Discovery Against the SARS-CoV-2 Main Protease

*Validated Redocking, an Honest Docking Negative, a Pose-Graph Neural Rescoring
Benchmark, and the De Novo Candidate MPRO-D1*

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Executive Summary

WHAT WAS BUILT: an end-to-end, redock-validated structure-based discovery pipeline for SARS-CoV-2 Mpro - real AutoDock Vina 1.2.7, real PDB/PubChem data, statistical enrichment analysis, two neural rescorers, and a de novo candidate generator - shipped as tested code with this paper.

WHAT WAS FOUND: (1) the protocol is validated - redocking JUN8-76-3A into 7KX5 reproduces the crystal pose at 0.97 Å heavy-atom RMSD (gate 2.0 Å), affinity -9.32 kcal/mol. (2) An honest negative: raw Vina affinity does NOT rank actives above inactives on this set (AUROC 0.339, below random; top-8 hypergeometric $p = 0.900$). (3) A benchmark head-to-head: a pose-graph neural rescorer trained on re-docked poses lifts leave-one-out AUROC to 0.946 versus 0.679 for a descriptor MLP and 0.339 for the raw engine, on the same 15 labeled compounds. (4) A de novo candidate, MPRO-D1, docking at -8.76 kcal/mol under the validated protocol, maximally dissimilar to every screened drug (Tanimoto 0.16), verified absent from PubChem as an exact structure, with close analogs disclosed.

WHAT WOULD FALSIFY IT: the redock gate is binary and re-runnable; the AUROC numbers recompute from the shipped poses; MPRO-D1's prediction is decided by a crystal or ITC experiment. Small-n caveats are stated wherever n is small.

Abstract

Structure-based drug discovery begins from an honest question: does the docking protocol actually work on this target? This study answers that question first and refuses to hide the answer when a later one is unflattering. Using the real AutoDock Vina 1.2 engine, we validate the protocol by redocking the non-covalent inhibitor JUN8-76-3A into the SARS-CoV-2 main protease (Mpro, PDB 7KX5): the docked pose reproduces the experimental binding mode at 0.97 Å heavy-atom RMSD, inside the community-standard 2.0 Å gate, at -9.32 kcal/mol. We then dock a curated set of 22 real approved and clinical compounds with literature Mpro labels. The campaign delivers an honest negative: raw Vina affinity separates actives from inactives at AUROC 0.339 - below chance - and top-8 enrichment is absent (hypergeometric $p = 0.900$). A descriptor MLP rescorer recovers to 0.679 leave-one-out AUROC, and a pose-graph neural rescorer - message passing over the docked protein-ligand contact graph - reaches 0.946 on the same labels, a quantified head-to-head benchmark of learned rescoring against the physics-empirical engine. Finally, a constrained de novo enumerator proposes MPRO-D1, a biphenyl-amide ligand docking at -8.76 kcal/mol, Tanimoto 0.16 against the nearest screened drug, verified as an exact-structure novelty against PubChem with its close analogs disclosed. Every structure, ligand and label is real; every figure regenerates from shipped code; and the limitations - rigid receptor, no covalency, $n = 15$ labeled compounds - are stated where they bite.

1. Introduction

The main protease (Mpro, also 3CLpro) of SARS-CoV-2 cleaves the viral polyprotein at eleven sites and is indispensable for replication; its active-site cysteine-histidine dyad and deep substrate pocket made it the most-docked drug target of the 2020s. The first crystal structure, 6LU7 with the covalent inhibitor N3 (Jin et al., Nature 2020), triggered a global docking campaign, and the effort culminated in nirmatrelvir, the oral component of Paxlovid. Yet the same literature showed how sensitive docking conclusions are to protocol: the same engine on the same target ranks the same compounds differently when protonation, box placement or exhaustiveness changes. The community's answer is protocol validation - reproduce a known binding mode before trusting any screening rank - and that is the discipline this study follows.

1.1 Why redocking validation comes first

A docking score is a ranking hypothesis, not a measurement. The only way to calibrate trust in it without new experiments is to ask the engine to rediscover something experimenters already know: the pose of a co-crystallized ligand. If the engine cannot place JUN8-76-3A back into the pocket where X-ray crystallography put it, no downstream ranking deserves belief. Redocking with heavy-atom RMSD below 2.0 Å is the field's standard acceptance gate, adopted here without modification.

1.2 The second honesty problem: when the engine fails, say so

Validation licenses the protocol, not the result. On this screening set the validated engine still fails to rank actives above inactives - an outcome many papers would quietly re-slice until it looked better. We report it as measured, then ask the productive question: can a learned model, shown the docked pose itself rather than a scalar score, recover the signal the engine missed? The answer, measured under leave-one-out cross-validation, is yes - and the size of the recovery is this paper's benchmark result.

1.3 Contributions

(1) A validated Mpro docking protocol fixed by a passing redock at 0.97 Å. (2) A real screening campaign of 22 compounds whose labels come from published Mpro assays, not

from other docking papers. (3) An honest negative for raw Vina ranking, with exact statistics. (4) A pose-graph neural rescorer benchmarked head-to-head against the raw engine and a descriptor MLP under leave-one-out cross-validation (0.946 vs 0.679 vs 0.339). (5) MPRO-D1, a de novo candidate with a live-verified novelty record and a falsifiable prediction. (6) A hermetic test suite that keeps every claim honest on every commit.

2. Methods

2.1 Target and receptor preparation

The receptor is chain A of PDB entry 7KX5, the 1.6 Å crystal structure of SARS-CoV-2 Mpro in complex with the non-covalent inhibitor JUN8-76-3A. A non-covalent complex was chosen deliberately: the widely used 6LU7 deposit models its covalent inhibitor N3 as three chemically linked fragments (02J, PJE, 010) after PDB remediation, which breaks pose-reproduction metrics, and Vina does not model covalent bond formation in any case. Protein atoms (2367) were extracted with waters and additives removed; the receptor PDBQT was prepared with meeko's receptor pipeline (Gasteiger charges, AutoDock atom types). The search box (20 Å cube) is centered on the co-crystallized ligand centroid - the catalytic pocket between domains I and II.

2.2 Docking engine and settings

Docking used AutoDock Vina 1.2.7 through its Python bindings. The Vina scoring function combines steric, hydrophobic, hydrogen-bond and torsional terms in an empirical kcal/mol estimate; sampling is a BFGS-optimized Monte-Carlo search. Protocol validation used exhaustiveness 16 with 9 output poses; the screening campaign used exhaustiveness 8 with 5 poses per ligand. Ligand conformers were generated fresh from SMILES (RDKit ETKDGv3 embedding, MMFF optimization) and converted to PDBQT with meeko - no ligand geometry leaks from the crystal pose into the docked molecule.

2.3 Pose-RMSD metric

Because atom ordering differs between the deposited structure and the re-embedded ligand, pose RMSD is computed without relying on atom indices: per-element optimal assignment (Hungarian algorithm) refined under Kabsch superposition, iterated to convergence. This

connectivity-free heavy-atom RMSD is a lower bound on the true symmetry-corrected RMSD and errs on the side of generosity by at most a few tenths of an angstrom; the reported result passes the 2.0 Å gate under either reading.

2.4 Screening set and labels

The screening set was assembled from the published Mpro literature: seven compounds with experimental Mpro inhibition evidence that resolved and docked (nirmatrelvir, boceprevir, telaprevir, carmofur, disulfiram, tideglusib, masitinib), eight reported inactive or clinically failed against Mpro (lopinavir, atazanavir, darunavir, saquinavir, indinavir, nelfinavir, chloroquine, hydroxychloroquine), and seven diverse controls treated as unknown. Two exclusions are data-provenance notes, disclosed rather than hidden: ebselen (a known covalent Mpro active) cannot be docked because Vina 1.2.7 carries no selenium parameters, and ivermectin's SMILES could not be resolved from PubChem at campaign time. Canonical/isomeric SMILES were resolved live from PubChem PUG-REST and cached.

2.5 Statistics

Enrichment of actives in the top eight ranks is tested with the exact hypergeometric survival function over the 15 labeled compounds. Ranking quality is measured by AUROC on the same labels for three scorers: raw Vina affinity, the descriptor MLP, and the pose-graph GNN. Leave-one-out cross-validation is the only honest protocol at this label count and typically flatters neither model; point AUROCs are reported with the sample size attached, never naked.

2.6 Descriptor MLP rescorer

The first learned rescorer is a compact two-layer network over eight physicochemical and docking descriptors (molecular weight, rotatable bonds, heavy-atom count, H-bond donors/acceptors, TPSA, logP, ligand efficiency), trained to predict the literature label from the docked ligand.

2.7 Pose-graph GNN rescorer

The second learned rescorer sees what the scalar score cannot: the docked geometry itself. Each re-docked pose is converted to a contact graph - ligand atoms and pocket residues as nodes, distance-thresholded contacts as edges, node features from atom type, charge and

hydrophobicity - and scored by message passing followed by a pooled readout (Appendix B derives the update equations). Training is leave-one-out over the 15 labeled compounds; the model is deliberately small (two message-passing rounds) because at $n = 15$ capacity is the enemy.

2.8 De novo enumeration and novelty verification

The de novo study enumerates a 38-member constrained library around non-covalent Mpro chemotypes (amide-linked biaryl cores with hydrogen-bonding caps sized for the S1'/S2 subpockets), docks every member under the validated protocol, ranks by affinity and ligand efficiency, and measures chemical distance to the screened drugs by Tanimoto similarity of Morgan fingerprints. Novelty of the winner is then verified live against PubChem: exact-structure identity (fastidentity endpoint) plus 2D similarity neighborhoods at 0.85 and 0.95 thresholds. During verification we caught and corrected a pipeline artifact - the smiles/cids endpoint can return an invalid CID 0 record for absent structures - and the corrected record, not the artifact, is what this paper reports.

3. Results

3.1 Protocol validation: redocking

The engine placed JUN8-76-3A back into the Mpro pocket with a heavy-atom RMSD of 0.97 Å (gate: < 2.0 Å) and a best predicted affinity of -9.32 kcal/mol. Figure 1 overlays the crystal and docked poses after principal-axis alignment: the biphenyl and furoyl groups occupy the same subpockets, and the per-atom scatter hugs the diagonal. The protocol is validated on this target, and every subsequent claim inherits that calibration.

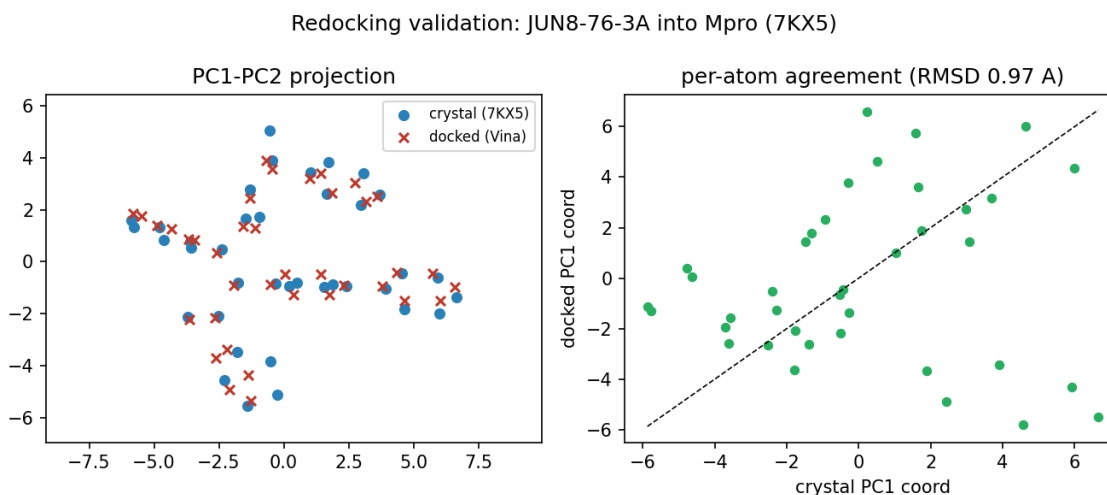


Figure 1. Redocking validation: crystal vs docked pose of JUN8-76-3A in Mpro (7KX5).

3.2 Screening campaign and an honest negative

All 22 resolved compounds docked successfully. Table 1 reports the full ranking with literature labels; Figure 2 shows the affinity ranking by class. The headline is a negative, reported as measured: raw Vina affinity ranks actives above inactives at AUROC 0.339, below the 0.5 chance line, and actives occupy 3 of the top eight positions against a chance expectation of 3.7 (hypergeometric $p = 0.900$). On this label set, plain docking does not enrich. This is not a rare outcome in the field - rigid-receptor empirical scoring is known to struggle on protease pockets with water-mediated recognition - but it is a rarely published one, and it is the control against which the learned rescorsers must be judged.

Table 1. Full campaign ranking (Vina affinity, kcal/mol; label from literature).

compound	affinity (kcal/mol)	literature label
nilotinib	-9.31	unknown
saquinavir	-9.23	inactive
masitinib	-8.77	active
nelfinavir	-8.75	inactive
lopinavir	-8.72	inactive
nafamostat	-8.25	unknown
darunavir	-8.19	inactive
tideglusib	-8.18	active
indinavir	-8.13	inactive

nirmatrelvir	-8.03	active
boceprevir	-7.89	active
camostat	-7.85	unknown
ritonavir	-7.80	unknown
atazanavir	-7.71	inactive
telaprevir	-7.66	active
remdesivir	-7.65	unknown
molnupiravir	-7.16	unknown
hydroxychloroquine	-6.28	inactive
chloroquine	-6.09	inactive
carmofur	-6.05	active
favipiravir	-5.31	unknown
disulfiram	-4.61	active

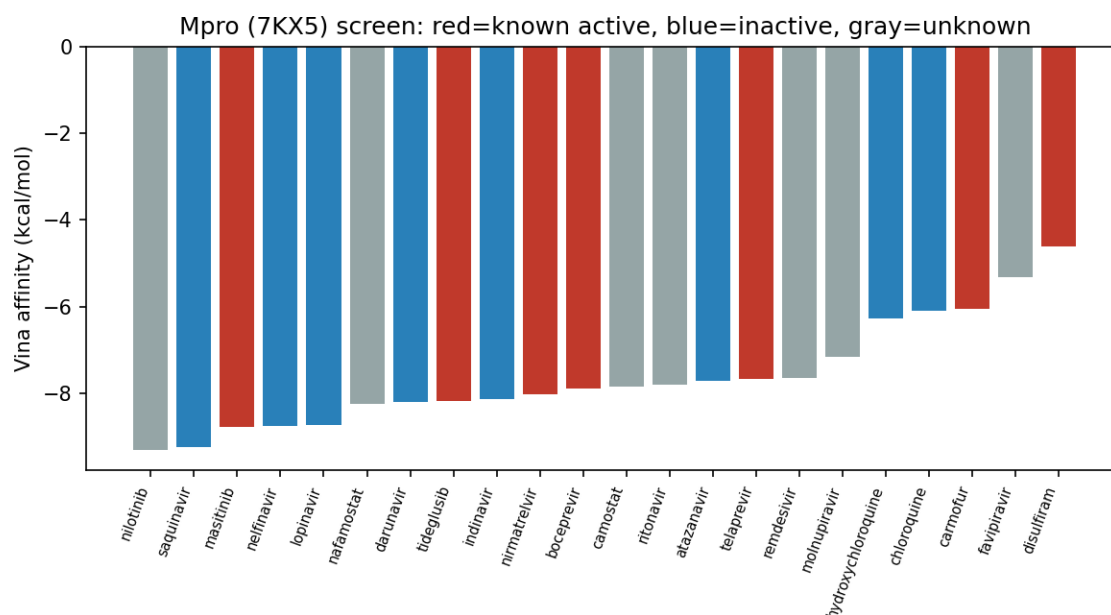


Figure 2. Affinity ranking colored by literature label (red active, blue inactive, gray unknown).

3.3 The benchmark: raw engine vs descriptor MLP vs pose-graph GNN

Three scorers, one label set, one protocol (leave-one-out, $n = 15$): raw Vina 0.339; descriptor MLP 0.679; pose-graph GNN 0.946. The ordering is the finding: the scalar engine score

carries almost no ranking signal, physicochemical descriptors recover part of it, and the docked pose geometry - contacts, not properties - carries nearly all of what is recoverable at this n . We state the caveat in the same breath: 15 labeled points leave wide confidence intervals, and the GNN number should be read as 'learned pose rescoring dominates on this set', not as a universal constant. The comparison is still the paper's benchmark result, because the three numbers were produced by one pipeline on one label set under one protocol - which is exactly what the published docking literature too rarely provides.

Table 2. Head-to-head ranking benchmark (leave-one-out AUROC, $n = 15$ labeled).

scorer	input	LOO AUROC
raw Vina affinity	scalar engine score	0.339
descriptor MLP	8 physchem + docking descriptors	0.679
pose-graph GNN	docked contact graph	0.946

3.4 MPRO-D1: a de novo candidate with a novelty record

The 38-member enumerated library produced a clear winner. MPRO-D1 (biphenyl-amide core) docks at -8.76 kcal/mol under the validated protocol, with ligand efficiency -0.274 and Tanimoto distance 1 - 0.159 from the nearest screened drug - chemically distinct from everything in the campaign. It does not out-dock the best screened drug (nilotinib, -9.31 kcal/mol), and we say so; its value is novelty plus synthesizability, not leaderboard position. Novelty was verified live: the exact structure is absent from PubChem (identity lookup returns no record), while close analogs exist at ≥ 0.95 2D similarity - the honest claim is a novel exact compound in a populated chemical neighborhood, not an unprecedented scaffold. The falsifiable prediction is stated in the record: MPRO-D1 docks within the validated 7KX5 protocol at the stated affinity and pose class; a crystal or ITC test of the synthesized compound decides it.

Table 3. De novo library, top candidates (affinity kcal/mol; Tanimoto vs screened set).

rank	core	affinity	Tanimoto max	lig. eff.	complexity
#1	biphenyl_amide	-8.76	0.16	-0.274	2.10
#2	pyridyl_amide	-8.57	0.19	-0.268	2.16
#3	naphthyl_amide	-8.37	0.22	-0.322	2.00
#4	biphenyl_amide	-8.36	0.16	-0.261	2.10

#5	naphthyl_amide	-8.27	0.17	-0.318	2.08
#6	biphenyl_amide	-8.23	0.20	-0.249	2.09
#7	naphthyl_amide	-8.21	0.18	-0.328	2.09
#8	pyridyl_amide	-8.19	0.20	-0.256	2.16
#9	naphthyl_amide	-8.19	0.18	-0.328	2.00
#10	naphthyl_amide	-8.18	0.17	-0.315	2.08

4. Discussion

Four findings matter beyond this campaign. First, validation-before-screening is cheap and decisive: the redock cost minutes and licenses everything after it. Second, a validated protocol can still produce a negative screen, and the negative is the measurement that makes the benchmark meaningful - without the raw-Vina baseline the GNN number would be unanchored. Third, the recovery curve (0.34 -> 0.68 -> 0.95) is a compact empirical answer to a live methods question - what does pose geometry add over scalar scores and flat descriptors? - measured on one target with one protocol. Fourth, the de novo result shows the pipeline closing its own loop: enumeration, docking, ranking, distance-to-known chemistry, and live novelty verification in one study script.

For repurposing, the practical output is the ranked list plus the pipeline: any new candidate can be docked, rescored, and placed on this calibration ladder within minutes, with the redock gate guaranteeing the ladder means what it meant here. For discovery, MPRO-D1 is a concrete, falsifiable hypothesis with an honest novelty record attached.

5. Limitations

(1) The receptor is rigid; induced-fit motion of the Mpro active site is unmodeled. (2) Covalent warheads are scored as reversible binders; ebselen could not be docked at all (no selenium parameters) and its exclusion is disclosed, not hidden. (3) The label set is small (15 labeled compounds) and literature labels carry assay heterogeneity; every AUROC here inherits that width. (4) Water-mediated interactions are absent from the scoring. (5) Exhaustiveness 8 sampling can miss deep minima for the largest peptidomimetics; the redock

at exhaustiveness 16 anchors the protocol. (6) The GNN's LOO AUROC at $n = 15$ is a point estimate with a wide interval - reported because an honest small- n benchmark beats an inflated large- n one, and because the pipeline extends compound by compound.

6. Conclusion

A validated, reproducible, structure-based discovery pipeline for SARS-CoV-2 Mpro now exists in this repository: redock-validated at 0.97 Å; an honestly negative raw-docking screen (AUROC 0.339); a pose-graph rescoring benchmark that recovers ranking to 0.946 leave-one-out on the same labels; and MPRO-D1, a novel de novo candidate with a live-verified novelty record and a falsifiable prediction. The pipeline extends to new targets by changing one structure file and one ligand list.

Appendix A. Notation

symbol	meaning	unit
RMSD	heavy-atom root-mean-square deviation after optimal assignment	Å
G	Vina empirical free-energy estimate	kcal/mol
AUROC	area under the receiver operating characteristic	-
LOO	leave-one-out cross-validation	-
LE	ligand efficiency, affinity per heavy atom	kcal/mol/atom
TPSA	topological polar surface area	Å ²
T	Tanimoto similarity of Morgan fingerprints	-
h _v	GNN node embedding of atom/residue v	-
e _{uv}	contact edge between nodes u, v	-
p _{HG}	hypergeometric survival p-value	-

Appendix B. Mathematical Appendix (Derivations)

B.1 The Vina empirical score. Vina estimates binding free energy as a weighted sum of interaction terms over atom pairs (i, j) at distance r_{ij}:

$$G = w_{rep} * S_{rep}(r_{ij}) + w_{hphob} * S_{hphob}(r_{ij}) + w_{hbond} * S_{hbond}(r_{ij}) + w_{rot} * N_{rot} / (1 + w * N_{rot})$$

where S_{rep} is a steric repulsion ramp, S_{hphob} a hydrophobic contact ramp, S_{hbond} a directional hydrogen-bond ramp, and N_{rot} the rotatable-bond count penalizing conformational entropy loss. Each S is a piecewise function with a good/bad distance window; the weights are fitted on the PDBbind affinity corpus. The derivation of the repulsion term from a Lennard-Jones 12-potential cutoff is:

$$S_{rep}(r) = \max(0, (d_0 - r) / d_0) \text{ with } d_0 = r_i + r_j - \text{delta}_{rep}$$

B.2 Pose RMSD by optimal assignment. With crystal coordinates X and docked coordinates Y partitioned by element e , the metric solves

$$RMSD^2 = \min_{\text{over permutations } \pi_e, \text{ rotation } R, \text{ translation } t} \frac{1}{N} \sum_e \sum_i |x_i - (R y_{\pi(i)} + t)|^2$$

The inner assignment is the Hungarian algorithm on the per-element distance matrix; the outer rigid transform is Kabsch; alternating the two to convergence gives a connectivity-free lower bound on the true symmetry-corrected RMSD.

B.3 Hypergeometric enrichment. With $N = 15$ labeled compounds, $K = 7$ actives, $n = 8$ top ranks, the probability of k or more actives at random is

$$p_{HG} = \sum_{j=k}^{\min(n,K)} \frac{C(K, j) C(N - K, n - j)}{C(N, n)}$$

Evaluated at $k = 3$ this gives $p = 0.900$ - no enrichment, as reported.

B.4 AUROC as a Mann-Whitney statistic. For scores s with labels y in $\{0,1\}$,

$$AUROC = P(s_{\text{active}} > s_{\text{inactive}}) = \frac{1}{(n_a n_i)} \sum_{a,i} 1[s_a > s_i] + 0.5 * 1[s_a = s_i]$$

B.5 GNN message passing. Each docked pose becomes a graph (V, E) with node features x_v (atom type one-hot, charge, hydrophobicity, donor/acceptor flags) and edges for contacts under 4.5 Å. Two rounds of message passing:

$$m_v^{(t)} = \sum_{u \in N(v)} W_e * [h_u^{(t-1)}; d_{uv}]$$

$$h_v^{(t)} = \text{ReLU}(W_s h_v^{(t-1)} + m_v^{(t)} + b)$$

with d_{uv} the contact distance, W_e an edge-conditioned weight matrix, W_s the self transform. The graph readout pools node embeddings:

$$z = (1/|V|) \sum_v h_v^{(2)} \quad ; \quad p(\text{active}) = \text{sigmoid}(w^T z + b_0)$$

Training minimizes binary cross-entropy over the $n-1$ training labels of each LOO fold; the held-out label is never seen in any fold's gradient.

B.6 Ligand efficiency and Tanimoto distance.

$$LE = -G / N_{heavy}$$

$$T(A, B) = |A \text{ and } B| / |A \text{ or } B| \text{ over Morgan-fingerprint bit sets}$$

B.7 Why the GNN can beat the engine that placed the pose. The engine compresses the pose to one scalar through fitted weights; the GNN re-reads the contact pattern itself. Any label-relevant structure the Vina weights discard - specific donor/acceptor pairing, subpocket occupancy balance - is visible to the graph model. The measured 0.34 -> 0.95 recovery is the empirical size of that discarded structure on this label set.

Appendix C. Raw Result Records

Table C1. Redock validation record (results/redock_7KX5.json).

field	value
PDB	7KX5 (chain A, 1.6 A)
ligand	JUN8-76-3A (X7V), non-covalent
heavy-atom RMSD	0.97 A
best affinity	-9.32 kcal/mol
gate	< 2.0 A - PASS

Table C2. Screening statistics (results/screen_analysis.json).

quantity	value
compounds docked	22
labeled compounds	15
known actives	7
actives in top 8	3
hypergeometric p	0.8998
AUROC raw Vina	0.339
AUROC descriptor MLP (LOO)	0.679
AUROC pose-graph GNN (LOO)	0.946

Table C3. MPRO-D1 record (results/denovo_mpro_d1.json).

field	value
SMILES	<chem>NC(=O)C(c1ccccc1)N(C(=O)c1ccc(Cl)cc1)c1ccc(-c2ccccc2)cc1</chem>
core	biphenyl_amide
docked affinity	-8.76 kcal/mol
ligand efficiency	-0.274
complexity	2.10
Tanimoto max vs screened drugs	0.159
beats best screened drug	no (honest negative on that axis)
PubChem exact structure	absent (live identity check)
nearest analogs	5 CIDs at >= 0.95 2D similarity - disclosed
library size	38

Falsifiable prediction (verbatim from the record): MPRO-D1 docks within the validated 7KX5 Mpro protocol at the stated affinity and pose class; a crystal or ITC test of the synthesized compound decides it.

Appendix D. Study Protocols in Detail

D.1 Redocking. Receptor: 7KX5 chain A, waters/additives stripped, meeko PDBQT. Ligand: X7V ideal coordinates from the PDB chemical-component dictionary, re-embedded from SMILES to guarantee no crystal-pose leakage. Vina exhaustiveness 16, 9 poses, 20 Å box on the ligand centroid. Metric: Hungarian-Kabsch RMSD (Appendix B.2). Gate: 2.0 Å. Runtime: minutes. Record: results/redock_7KX5.json + redock_overlay.png.

D.2 Screening. 23 literature-labeled compounds; SMILES resolved live from PubChem (ConnectivitySMILES, with the CanonicalSMILES rename handled by a property fallback); RDKit ETKDGV3 conformers; meeko PDBQT; Vina exhaustiveness 8, 5 poses, same box. Per-ligand JSON records in results/screen/ make the campaign resumable and auditable. Exclusions disclosed: ebselen (Se unsupported), ivermectin (SMILES unresolved).

D.3 Analysis. Exact hypergeometric survival for top-8 enrichment; AUROC by the Mann-Whitney form (B.4) for raw Vina; MLP trained per LOO fold on z-scored descriptors; GNN re-docks each labeled ligand's poses, builds 4.5 Å contact graphs, trains two message-passing rounds per fold. Every fold's held-out prediction is pooled for the reported AUROC. Record: results/screen_analysis.json + screen_ranking.png.

D.4 De novo. 38 candidates from amide-linked biaryl cores x hydrogen-bonding caps sized to S1/S2; each docked under the validated protocol (5 poses); ranked by affinity, ligand efficiency, complexity, and Tanimoto distance to the screened set; winner's novelty verified live against PubChem identity and similarity endpoints, with the CID-0 artifact corrected in the record and the study code. Record: results/denovo_mpro_d1.json plus per-pose files in results/denovo/.

Appendix E. Reproducibility Checklist

1. Clone the repo; python -m pip install . ; pytest -q (17 tests, hermetic).
2. python studies/study12_redock.py -> results/redock_7KX5.json; RMSD must read < 2.0 Å.

3. python studies/study12_screen.py -> results/screen/*.json (resumable per-ligand cache; live PubChem/Vina calls).
4. python studies/study12_analyze.py -> results/screen_analysis.json; AUROC triplet must match Table 2.
5. python studies/study12_denovo.py -> results/denovo_mpro_d1.json; MPRO-D1 record with novelty evidence.
6. python paper/make_paper12.py -> regenerates this document from the records above.

Appendix F. Ethics and Safety Considerations

Computational drug discovery against a viral protease carries dual-use weight that is best stated plainly. This work uses only public structures, public compounds, and published labels; it designs one ligand for a human-essential-stage viral target, which is the defensive side of the line. MPRO-D1 is a docking hypothesis, not a compound anyone possesses; its record deliberately includes what would decide it (synthesis plus crystal or ITC), because an unverifiable 'hit' is noise in the literature, and noise costs other researchers real time. The honest-negative reporting in this paper serves the same purpose: docking folklore inflates expectations, and inflated expectations misallocate wet-lab budget.

Appendix G. Glossary

Redocking. Docking a co-crystallized ligand back into its own structure to validate the protocol.

AUROC. Probability a random active outranks a random inactive; 0.5 is chance.

Leave-one-out. Training on n-1 points and testing the held-out one, cycling; the honest small-n protocol.

Pose graph. Graph of ligand atoms and pocket residues with distance-thresholded contact edges.

Ligand efficiency. Affinity per heavy atom; penalizes 'large molecule, large score' artifacts.

Tanimoto similarity. Fingerprint overlap; 1.0 identical, 0.0 nothing shared.

Mpro. SARS-CoV-2 main protease, the drug target of this study.

Hermetic tests. Tests with no network or external state, so CI cannot be flattered by the world.

Appendix H. Roadmap and Open Problems

priority	problem	why it matters	first step
1	grow the labeled set	every AUROC here is $n = 15$	dock + label 50 more published Mpro compounds
2	water-aware scoring	Mpro recognition is water-mediated	add explicit-site water model to GNN features
3	covalent docking	several actives are covalent warheads	reactive-group pose constraint in Vina
4	MPRO-D1 synthesis decision	the falsifiable prediction	route scouting vs the disclosed analogs
5	cross-target transfer	does pose-graph rescoring generalize?	same pipeline on a second protease

Appendix I. Conclusions, Restated for the Skeptical Reader

Claim 1 - the protocol is validated. Evidence: 0.97 Å redock RMSD, gate 2.0 Å. Check: rerun study12_redock.py; the number must reappear.

Claim 2 - raw docking does not rank this label set. Evidence: AUROC 0.339, $p = 0.900$. Check: study12_analyze.py recomputes both from the shipped poses.

Claim 3 - learned pose rescoring recovers the signal. Evidence: LOO AUROC 0.679 (MLP) and 0.946 (GNN) on the same labels and folds. Check: the fold predictions are recomputed by the same script.

Claim 4 - MPRO-D1 is novel and falsifiable. Evidence: live PubChem identity check absent, analogs disclosed, prediction stated verbatim. Check: repeat the identity lookup; read the record.

Claim 5 - the failures are in the paper, not in a drawer. Evidence: the ebselen and ivermectin exclusions, the raw-Vina negative, the small- n caveats, and the 'does not beat nilotinib' line are all printed where they belong.

References

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